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# 🧠 EDA Interview Questions & Answers

#### 1. What is EDA and why is it important
**Answer:** Exploratory Data Analysis (EDA) is the process of understanding, cleaning, and visualizing data before building ML models. It helps identify missing values, outliers, patterns, and relationships.

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#### 2. Difference between `info()` and `describe()`
**Answer:** `info()` shows dataset structure, data types, and non-null counts, while `describe()` provides statistical summaries such as mean, median, standard deviation, and quartiles.

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#### 3. Difference between mean, median, and mode
**Answer:** Mean is affected by outliers, median is robust to outliers, and mode represents the most frequent value. Mean can mislead in skewed data, while mode may not represent continuous data well.

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#### 4. Why do we scale data? Name two algorithms that don't require scaling, while Decision Trees and Random Forests do.
**Answer:** Scaling ensures features have the same range. Algorithms that don't require scaling are Linear Regression and Logistic Regression. Algorithms that do require scaling are Decision Trees and Random Forests.

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#### 5. What is the IQR and how is it used?
**Answer:** IQR (Interquartile Range) =  $Q3 - Q1$ . Data points above  $Q3 + 1.5 \times IQR$  and below  $Q1 - 1.5 \times IQR$  are considered outliers.

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#### 6. What are outliers? Should you always remove them?
**Answer:** Outliers are unusually large or small observations. They should be removed if they are errors or irrelevant, but not if they are important data points.

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#### 7. Difference between StandardScaler and MinMaxScaler
**Answer:** StandardScaler standardizes data by subtracting the mean and dividing by the standard deviation. MinMaxScaler rescales values between 0 and 1.

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🧠 EDA Interview Questions & Answers

1. What is EDA and why is it important before Machine Learning?

Answer: Exploratory Data Analysis (EDA) is the process of understanding, cleaning, and visualizing data before building ML models. It helps identify missing values, outliers, patterns, and relationships.

2. Difference between `info()` and `describe()`?

Answer: `info()` shows dataset structure, data types, and non-null counts, while `describe()` provides statistical summaries such as mean, median, standard deviation, and quartiles.

3. Difference between mean, median, and mode — when would each mislead you?

Answer: Mean is affected by outliers, median is robust to outliers, and mode represents the most frequent value. Mean can mislead in skewed data, while mode may not represent continuous data well.

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### 8. What is the difference between Label Encoding and One-Hot Encoding?
**Answer:** Label Encoding assigns numbers to categories. One-Hot Encoding creates binary columns. Label Encoding is used for tree-based models where numeric order is not important.

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### 9. What does a correlation coefficient indicate?
**Answer:** It indicates a very strong negative correlation between two variables.

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### 10. What's the difference between correlation and causation?
**Answer:** Correlation means two variables move together. Causation means one variable directly causes the other.

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### 11. What is the difference between a Series and a DataFrame?
**Answer:** A Series is a one-dimensional array. A DataFrame is a two-dimensional table with rows and columns.

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### 12. What does `df.shape` return, and how do you get the total number of elements?
**Answer:** `df.shape` returns (rows, columns). Total number of elements is df.size or df.count().

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### 13. Explain the difference between structured and unstructured data.
**Answer:** Structured data has a fixed format (e.g., CSV, SQL). Unstructured data has no predefined format (e.g., text, images).

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### 14. What is the difference between a feature and a target variable?
**Answer:** Features are input variables used for prediction. The target variable is the output the model predicts.

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### 15. What are the different types of missing data (MCAR, MAR, MNAR)?
**Answer:**
- MCAR: Missing completely at random.
- MAR: Missing depends on other variables.
- MNAR: Missing depends on the missing value itself.

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4. Why do we scale data? Name two algorithms that need it and two that don't.

Answer: Scaling ensures features have similar ranges. KNN and K-Means require scaling, while Decision Trees and Random Forest do not.

5. What is the IQR and how is it used to detect outliers?

Answer: IQR (Interquartile Range) = $Q3 - Q1$. Values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are considered outliers.

6. What are outliers? Should you always remove them?

Answer: Outliers are unusually large or small values. They should only be removed if they are errors or irrelevant, not if they represent valid observations.

7. Difference between StandardScaler and MinMaxScaler?

Answer: StandardScaler standardizes data using mean and standard deviation, while MinMaxScaler rescales values between 0 and 1.

8. What is the difference between Label Encoding and One-Hot Encoding? When would Label

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- MNAR: Missing depends on the missing value
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### 16. Why might forward-fill be a poor choice?
**Answer:** Because previous values may not be representative of the current or misleading data.
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### 17. What is data leakage, and how can it be avoided?
**Answer:** Data leakage occurs when information from the test set is used in the training set, leading to overly optimistic model performance.
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### 18. Explain Z-score and how it's used in outlier detection.
**Answer:** Z-score measures how many standard deviations a data point is from the mean. Values with  $|Z| > 3$  are commonly considered outliers.
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### 19. What's the difference between a histogram and a bar chart?
**Answer:** Histograms show the distribution of continuous data, while bar charts compare categorical data.
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### 20. When would you use a violin plot instead of a box plot?
**Answer:** Use a violin plot when you want to visualize the full distribution shape of a continuous variable.
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### 21. What is the purpose of a pairplot?
**Answer:** A pairplot visualizes the relationship between multiple variables and their distributions.
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### 22. Why is `nunique()` useful when doing exploratory data analysis?
**Answer:** It identifies columns with only one unique value, which may not be useful information for prediction.
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### 23. What's the risk of using a pie chart for data visualization?
**Answer:** Too many slices make the chart difficult to read and interpret accurately.

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Encoding actually be correct to use on nominal data?

Answer: Label Encoding assigns numbers to categories, while One-Hot Encoding creates binary columns. Label Encoding can be used on nominal data for tree-based models where numeric order is not interpreted.

9. What does a correlation coefficient of -0.9 mean?

Answer: It indicates a very strong negative linear relationship between two variables.

10. What's the difference between correlation and causation?

Answer: Correlation means two variables move together, while causation means one variable directly causes the other.

11. What is the difference between a Series and a DataFrame in Pandas?

Answer: A Series is a one-dimensional data structure, while a DataFrame is a two-dimensional table with rows and columns.

12. What does `df.shape` return, and how is it different from `df.size`?

Answer: `df.shape` returns (rows, columns), whereas `df.size` returns the total number of elements.

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### 24. Explain `groupby()` in your own words.
**Answer:** `groupby()` groups similar data for
calculations. For example, finding the average
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### 25. What is the difference between `pivot` and `groupby`?
**Answer:** `groupby()` performs grouped calculations, while
`pivot` summarizes data in a spreadsheet-like table.
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### 26. Why should feature scaling be fit on the training set, not the full dataset?
**Answer:** Fitting on the full dataset causes data leakage from the test set, influencing preprocessing.
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### 27. What's the difference between discrete and continuous data?
**Answer:** Discrete data consists of countable values (e.g., number of students), while continuous data can take any value within a range (e.g., height).
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### 28. What is Ordinal Encoding, and how is it different from Label Encoding?
**Answer:** Ordinal Encoding preserves the order of categories, while Label Encoding simply assigns numbers to categories.
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### 29. Give one example each of nominal, ordinal, and binary variables.
**Answer:**
- Nominal: Color (Red, Blue, Green)
- Ordinal: Education Level (High School, College, Graduate)
- Binary: Purchased (Yes/No)
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### 30. Walk through the complete EDA pipeline and explain it in an interview.
**Answer:**
1. Load the dataset.
2. Understand its structure (`shape`, `info`, `dtypes`).
3. Check missing values and duplicates.
4. Handle missing values and remove duplicates.
5. Detect and treat outliers.
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13. Explain the difference between structured and unstructured data with examples.

Answer: Structured data has a fixed format (e.g., Excel tables), while unstructured data has no predefined format (e.g., images, videos, emails).

14. What is the difference between a feature and a target variable?

Answer: Features are input variables used for prediction, while the target variable is the output the model predicts.

15. What are the different types of missing data mechanisms (MCAR, MAR, MNAR)?

Answer:

- MCAR: Missing completely at random.
- MAR: Missing depends on other variables.
- MNAR: Missing depends on the missing value itself.

16. Why might forward-fill be a poor choice for a randomly-ordered dataset?

Answer: Because previous values may not be related, resulting in incorrect or misleading data.

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6. Perform univariate analysis.  
7. Perform bivariate and multivariate anal  
8. Visualize data using plots.  
9. Encode categorical variables if needed.  
10. Scale numerical features if required.  
11. Identify important features using corr  
12. Prepare the cleaned dataset for Machin
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17. What is data leakage, and how can improper preprocessing cause it?

Answer: Data leakage occurs when information from the test set influences training, leading to overly optimistic model performance.

18. Explain Z-score and how it's used for outlier detection.

Answer: Z-score measures how many standard deviations a value is from the mean. Values with $|Z| > 3$ are commonly considered outliers.

19. What's the difference between a histogram and a bar chart?

Answer: Histograms show the distribution of continuous numerical data, while bar charts compare categorical data.

20. When would you use a violin plot instead of a boxplot?

Answer: Use a violin plot when you want to see both summary statistics and the full distribution shape.

21. What is the purpose of a pairplot?

Answer: A pairplot visualizes relationships between multiple numerical variables and their distributions.

22. Why is `nunique()` useful when deciding which columns to drop before modelling?

Answer: It identifies columns with only one unique value, which provide no useful information for prediction.

23. What's the risk of using a pie chart with 10+ categories?

Answer: Too many slices make the chart difficult to read and compare accurately.

24. Explain `groupby()` in your own words, using a real example.

Answer: `groupby()` groups similar data together before applying calculations. For example, finding the average salary for each department.

25. What is the difference between `pivot_table()` and `groupby()`?

Answer: `groupby()` performs grouped calculations, while `pivot_table()` summarizes data in a spreadsheet-like table with rows and columns.

26. Why should feature scaling be fit only on the training data, not the full dataset?

Answer: Fitting on the full dataset causes data leakage because information from the test set influences preprocessing.

27. What's the difference between discrete and continuous numerical data?

Answer: Discrete data consists of countable values (e.g., number of students), while continuous data can take any value within a range (e.g., height).

28. What is Ordinal Encoding, and how is it different from Label Encoding in practice?

Answer: Ordinal Encoding preserves the natural order of categories, while Label Encoding simply assigns numbers and is mainly used for target labels.

29. Give one example each of nominal, ordinal, and binary categorical variables.

Answer:

- Nominal: Color (Red, Blue, Green)
 - Ordinal: Education Level (High School, Bachelor's, Master's)
 - Binary: Purchased (Yes/No)
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30. Walk through the complete EDA pipeline, step by step, as you would explain it in an interview.

Answer:

1. Load the dataset.
2. Understand its structure (`shape`, `info()`, `describe()`).
3. Check missing values and duplicates.

4. Handle missing values and remove duplicates.
5. Detect and treat outliers.
6. Perform univariate analysis.
7. Perform bivariate and multivariate analysis.
8. Visualize data using plots.
9. Encode categorical variables if needed.
10. Scale numerical features if required.
11. Identify important features using correlation and domain knowledge.
12. Prepare the cleaned dataset for Machine Learning.